

Network Analysis on Rumination

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2. Methods

2.1 Network analysis

2.1.1 Removing redundant items

As a preparation step, we applied the *goldbricker* function from the R package *networktools* to search for pairs of items in the RRS, SCRS and MHS items which were highly correlated ($r > 0.50$) and showed highly the same pattern with other items. When such “bad pairs” were identified, reduction techniques were put into use (with the *net_reduce* function in R): instead of the pair, the first principal component of the two variables were kept in the data as a new, merged variable.

The RRS items required no reduction steps as the highly correlating items did not extend their pattern to more nodes (their correlations to other nodes did not violate the criteria that $> 75\%$ of these correlations did not differ significantly). Among the SCRS scales the SCRS_8 & SCRS_1 scales required reductions steps using PCA. Among the MHC nodes MHC_3 & MHC_1 items were subject to reduction.

2.1.2 Network Analysis with LASSO

For the network analysis, the tuning parameter of *gamma* in the graphical LASSO algorithm has been set to 0.5 value in accordance with Bernstein et al in order to achieve sparse graphs. The final networks were chosen by the lowest values of the *extended Bayesian Information Criterion* (EBIC).

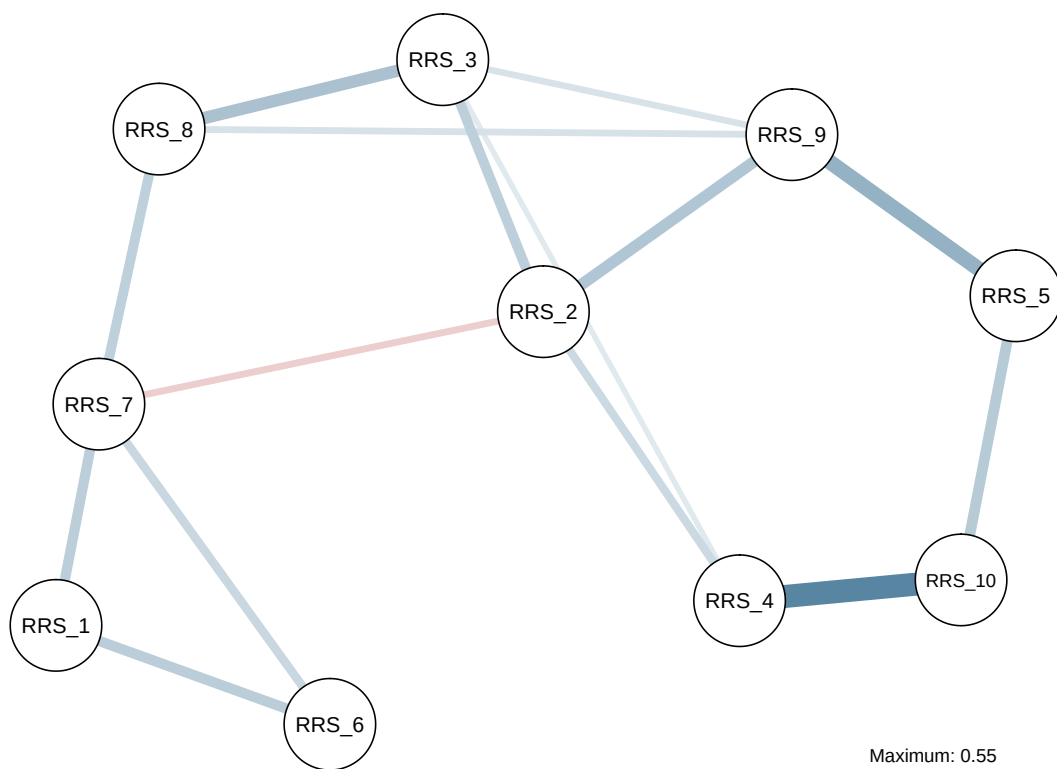
2.1.3. Community detection

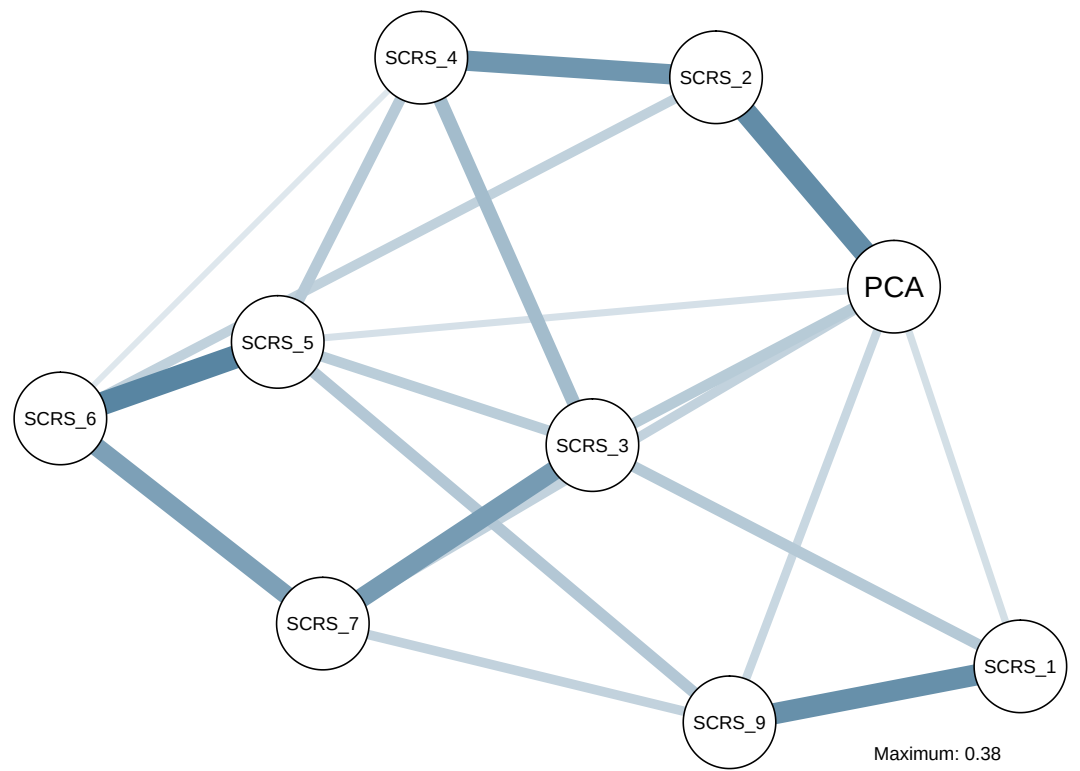
In order to get an understanding of community of nodes and to see if nodes can be grouped into neighbourhoods, we used the *igraph* R package. When detecting possible communities of nodes, the detection happened with the following parameters in the *spinglass algorithm*: TODO: try the walktrap algorithm from igraph!

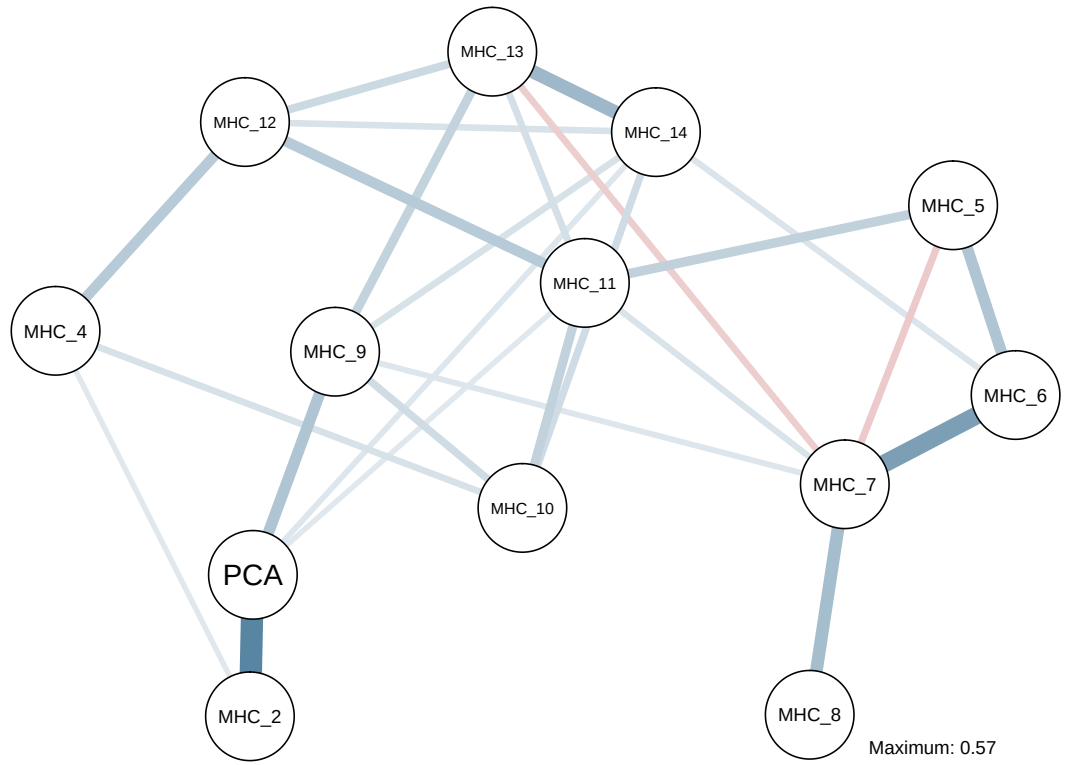
3. Results

3.1 Network plots

RSS network





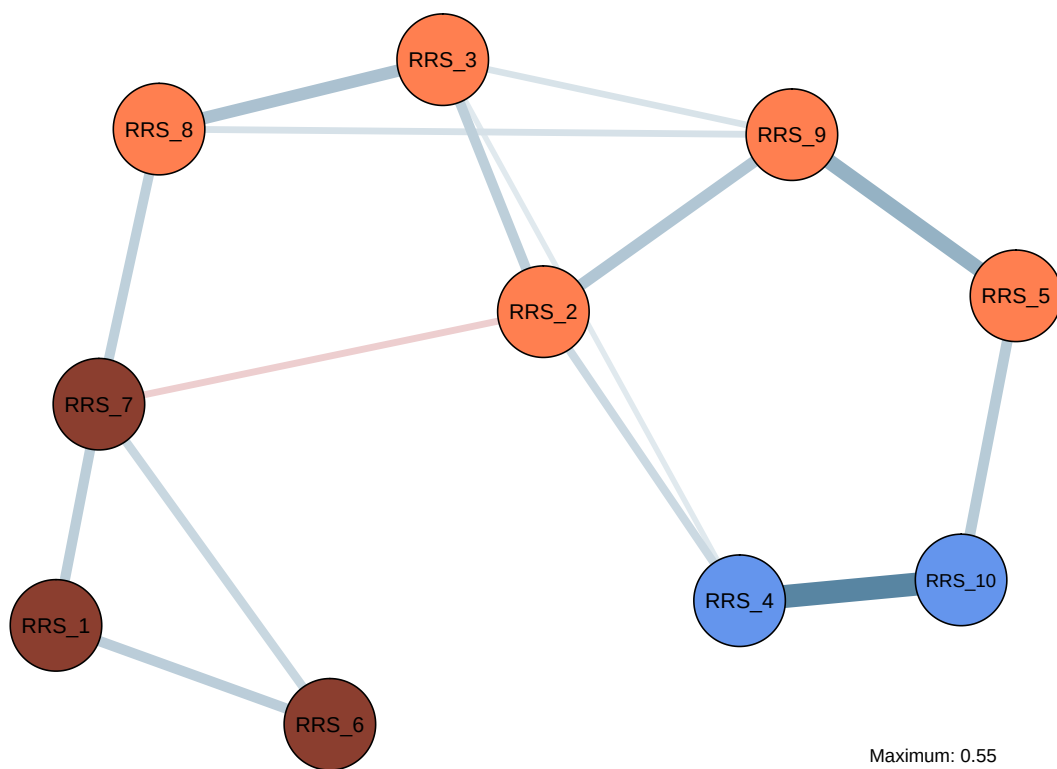


MHC network

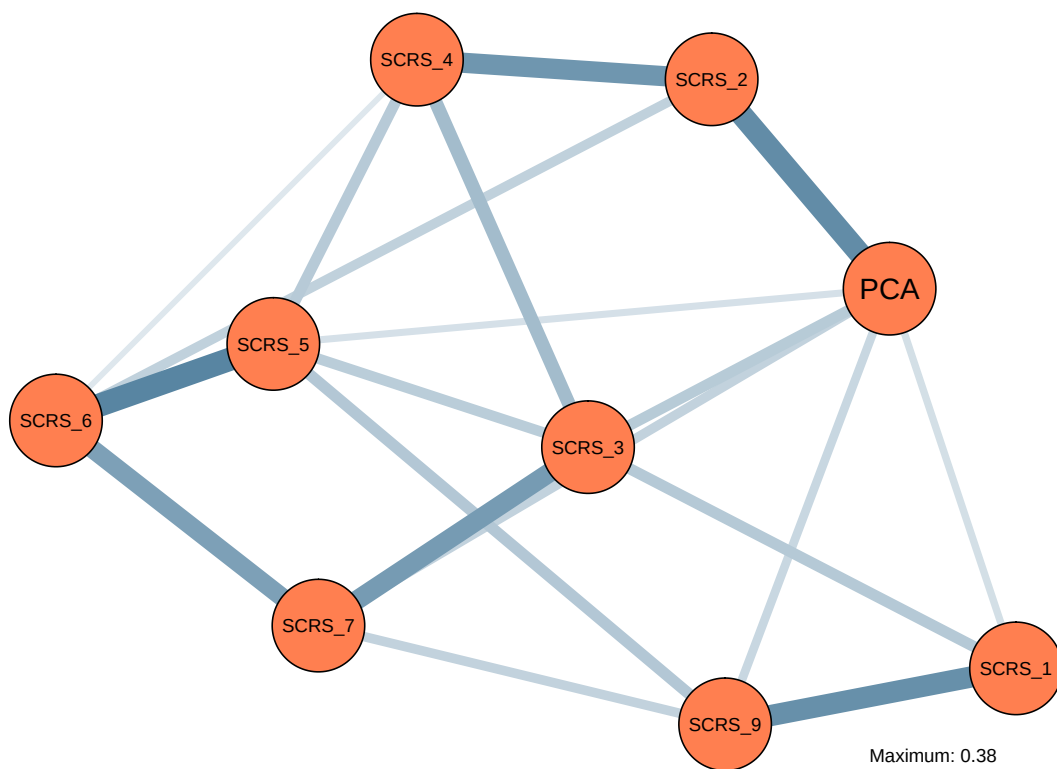
3.2 Community detection

Community mapping of RSS, MHC and SCRS with the *spinglass* algorithm:

RSS communities identified with parameters $\gamma = 0.5$, $\text{spins} = 25$, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99.



SCRS communities identified with parameters $\gamma = 0.5$, $\text{spins} = 25$, $\text{starting temperature} = 1$, $\text{ending temperature} = 0.01$, $\text{cooling factor} = 0.99$.



MHC communities identified with parameters $\gamma = 0.5$, $\text{spins} = 25$, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99.

